Assignment 8 Solutions

1.1 $\Gamma = \{x : \bot \to \bot, y : \bot, z : \bot \to \bot\}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash z : \bot \to \bot}\,\text{var} \quad \dfrac{\dfrac{}{\Gamma \vdash x : \bot \to \bot}\,\text{var} \quad \dfrac{}{\Gamma \vdash y : \bot}\,\text{var}}{\Gamma \vdash xy : \bot}\,\text{app}}{\Gamma \vdash z(xy) : \bot}\,\text{app}}{\{x : \bot \to \bot, y : \bot, z : \bot \to \bot\} \vdash z(xy) : \bot}\,\text{app}}{\{x : \bot \to \bot, y : \bot\} \vdash \lambda z.\, z(xy) : (\bot \to \bot) \to \bot}\,\text{fun}$$

1.2 $\Gamma = \{f : \bot \to \bot, x : \bot\}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash f : \bot \to \bot}\,\text{var} \quad \dfrac{\dfrac{}{\Gamma \vdash f : \bot \to \bot}\,\text{var} \quad \dfrac{}{\Gamma \vdash x : \bot}\,\text{var}}{\Gamma \vdash fx : \bot}\,\text{app}}{\Gamma \vdash f(fx) : \bot}\,\text{app}}{\{f : \bot \to \bot, x : \bot\} \vdash f(fx) : \bot}\,\text{app}}{\{f : \bot \to \bot\} \vdash \lambda x.\, f(fx) : \bot \to \bot}\,\text{fun}}{\emptyset \vdash \lambda f.\, \lambda x.\, f(fx) : (\bot \to \bot) \to \bot \to \bot}\,\text{fun}$$

1.3 $\Gamma = \{x : \bot \to \bot \to \bot, y : \bot \to \bot, z : \bot\}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash x : \bot \to \bot \to \bot}\,\text{var} \quad \dfrac{}{\Gamma \vdash z : \bot}\,\text{var}}{\Gamma \vdash xz : \bot \to \bot}\,\text{app} \quad \dfrac{\dfrac{}{\Gamma \vdash y : \bot \to \bot}\,\text{var} \quad \dfrac{}{\Gamma \vdash z : \bot}\,\text{var}}{\Gamma \vdash yz : \bot}\,\text{app}}{\Gamma \vdash (xz)(yz) : \bot}\,\text{app}}{\{x : \bot \to \bot \to \bot, y : \bot \to \bot, z : \bot\} \vdash (xz)(yz) : \bot}\,\text{fun}}{\{x : \bot \to \bot \to \bot\} \vdash \lambda y.\, \lambda z.\, (xz)(yz) : (\bot \to \bot) \to \bot \to \bot}\,\text{fun}}{\emptyset \vdash \lambda x.\, \lambda y.\, \lambda z.\, (xz)(yz) : (\bot \to \bot \to \bot) \to (\bot \to \bot) \to \bot \to \bot}\,\text{fun}$$

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$$\dfrac{\dfrac{}{\{x : \bot\} \vdash x : \bot}\,\text{var}}{\emptyset \vdash \lambda x.\, x : \bot \to \bot}\,\text{fun} \qquad \dfrac{\dfrac{}{\{x : \bot \to \bot\} \vdash x : \bot \to \bot}\,\text{var}}{\emptyset \vdash \lambda x.\, x : (\bot \to \bot) \to \bot \to \bot}\,\text{fun}$$

5.1 $\Gamma = \{x : \bot, y : \bot \to \bot\}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\Gamma \vdash y : \bot \to \bot}\,\text{var} \quad \dfrac{}{\Gamma \vdash x : \bot}\,\text{var}}{\Gamma \vdash yx : \bot}\,\text{app}}{\{x : \bot, y : \bot \to \bot\} \vdash yx : \bot}\,\text{app}}{\{x : \bot\} \vdash \lambda y^{\bot \to \bot}.\, yx : (\bot \to \bot) \to \bot}\,\text{fun}}{\emptyset \vdash \lambda x^{\bot}.\, \lambda y^{\bot \to \bot}.\, yx : \bot \to (\bot \to \bot) \to \bot}\,\text{fun}$$

5.2 $\Gamma = \{ y : (\top \to \top) \to \top,\ x : \top \to \top \}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\top \to \top : x \vdash \Gamma}\,\text{var}}{\Gamma \vdash y : (\top \to \top) \to \top}\,\text{var}}{\Gamma \vdash y\ x : \top}\,\text{app}}{\top \to \top : x \vdash \Gamma}\,\text{var}}{\Gamma \vdash (y\ x) : \top}\,\text{app}}{\top \to (\top \to \top) : (y\ x) \vdash \{ y : (\top \to \top) \to \top \}}\,\text{fun}$$

5.3 $\Gamma = \{ w : \top \to \top \to \top,\ y : (\top \to \top) \to \top \}$

$\Gamma' = \{ w : \top \to \top \to \top,\ y : (\top \to \top) \to \top,\ x : (\top \to \top) \to \top \}$

$\Gamma'' = \{ w : \top \to \top \to \top,\ y : (\top \to \top) \to \top,\ x : (\top \to \top) \to \top,\ z : \top \}$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{\top : z \vdash \Gamma''}\,\text{var}}{\top \to \top : w\ z \vdash \Gamma''}\,\text{app}}{\top \to (\top \to \top) : x \vdash \Gamma''}\,\text{var}}{\top : (w\ z)\ x \vdash \Gamma''}\,\text{app}}{\top \to \top : \lambda z^{\top}.\ (w\ z)\ x \vdash \Gamma'}\,\text{fun}}{\top : (\lambda z^{\top}.\ (w\ z)\ x)\ x \vdash \Gamma'}\,\text{app}}{\top \to (\top \to \top) : \lambda x^{\top \to (\top \to \top)}.\ (\lambda z^{\top}.\ (w\ z)\ x)\ x \vdash \{ w : \top \to \top \to \top,\ y : (\top \to \top) \to \top \}}\,\text{fun}$$

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$$\dfrac{}{\top : x \vdash \{ x : \top,\ y : \top \to \top \}}\,\text{var}$$